A Web-Scale Analysis of Trends in AI-Generated and SEO Spam Content Using Human Annotation and Machine Learning

Master's Thesis Final Presentation

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Why do Online Reviews Matter?



Reviews are one of the most influential forms of online information.

Source: MarTech Zone — Statistics on the impact of online reviews on consumer behaviour

Compare The Best Propan

Showing 1 - 10 in 11 results

RANKING LIST | SORT BY SCORES

1 TOP CHOICE



Sleepingo Camping Sleeping Pad (Large) - Ultralight 14 OZ, Best Sleeping Pads for Camping, ding - Lightweight, pact, Camping Air

SCORE 9.8 AI Score

Brand Sleepingo

Prime

1 Bernzomatic Worth

1, 2, and 3. That's

7. Best In Quick Heating: Royal Massage Clamshell H Heater

4.2



BUY ON AMAZ

Image: *Royal Massage*

The Royal Massage Clamshell Hot Stone Heater is portable and lightweight. It car small stones or four large stones. It has a cool touch outer clamshell housing that handle and also safe for use. It also features dual heating pads, which heat up sto 122 degrees in 10 minutes. It is suitable for heating stones for massage, manicure facials, or any other smaller volumes.

Item Weight: 2.6 Pounds | **Color**: Blue | **Material**: Steel

Pros	Cons
Dual heating plates	None
Versatile heating system	
High Quality stones	
Lightweight	
Portable	

Our Experience



While it doesn't perfectly match th an extremely pleasant smell. We

Longevity: 4 out of 5

The Chemical Guys New Car Sme last for quite awhile, especially if d

Odor Masking: 4.5 out of 5

This may be one of the best car ai Other car air fresheners gradually bottle can be used to immediately

Value: 4 out of 5

You'll pay around $10 for 16.0 fluic diluted to last longer. You may run necessary. A couple of sprays sho

RMS Health 26" Long Grabber Reacher w/ Magnetic Tip

This is not a great reacher grabber. It looks like the Sammons model with a single hooked claw meeting a straight, smooth one with a magnetic tip, but the handle felt flimsy and the claw pickup tool portion failed to pick up anything other than soft, easy-to-grab items like stuffed animals. It might be useful for picking trash up off the ground, but probably not much else. If you want a grabber reacher tool with a curved claw, try the Sammons version instead.

Pros + Can pick up soft, light objects + Has magnetic tip

Cons — Claw doesn't hold onto anything other than light, soft items

— Handle feels flimsy and is hard to hold onto

Buy now at Amazon

$29.99 from Walmart

Two Core Problems in Review Pages



Problem 1: **Why was it written?**

- **Good review:** honest feedback, based on real use, pros and cons
- **Spam review:** tries to promote, template phrases, too positive, repetitive words



Problem 2: **Who wrote it?**

- **Human-written:** personal details, real experiences, subjective opinion
- **AI-written:** fluent, polished, no real experience, generic statements

Each problem captures different aspects of web content.

Research Questions

1. How accurately can models **identify review** pages and **classify them into good or spam** using annotated datasets?

2. How well can machine learning models **distinguish AI-generated text** from **human text** ?

3. What is the **correlation between AI-generated content and spam** review predictions?

4. How do AI-generated and spam review contents **evolve over the two-year period** across **different search engines**?

Overview of the Presentation

1. Dataset and Extraction

a. Built Annotation Interface

b. PAN Dataset

2. Methodology

3. Results

4. Conclusion

Dataset & Data Extraction

Extracted Main Readable Content

- Used resiliparse
- Removed boilerplate elements

Metadata for Each Page

- **Timestamp:** crawl month
- **Search engine:** Bing, DuckDuckGo, Startpage
- **URL:** original page link
- **File path:** WARC* file location to traceback

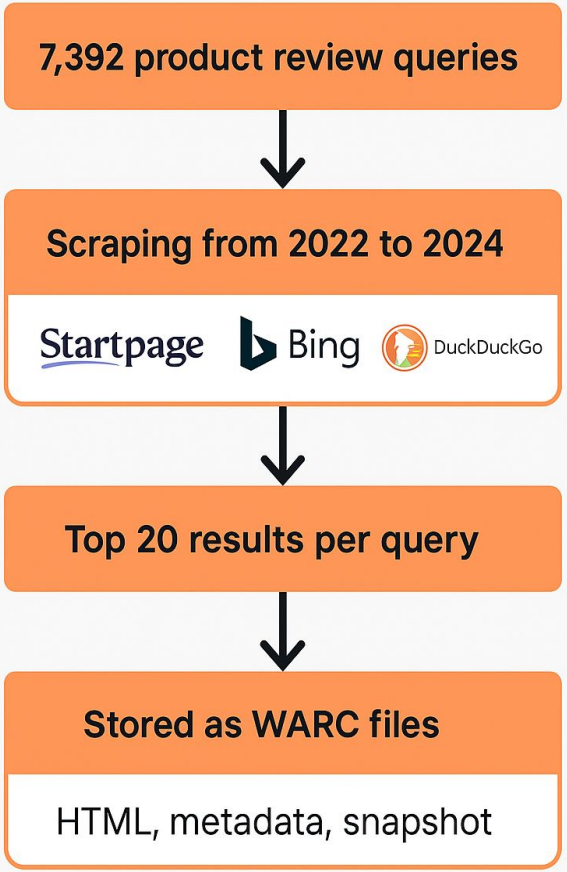
Used for both AI detection and Spam detection models

Complete webpage snapshot

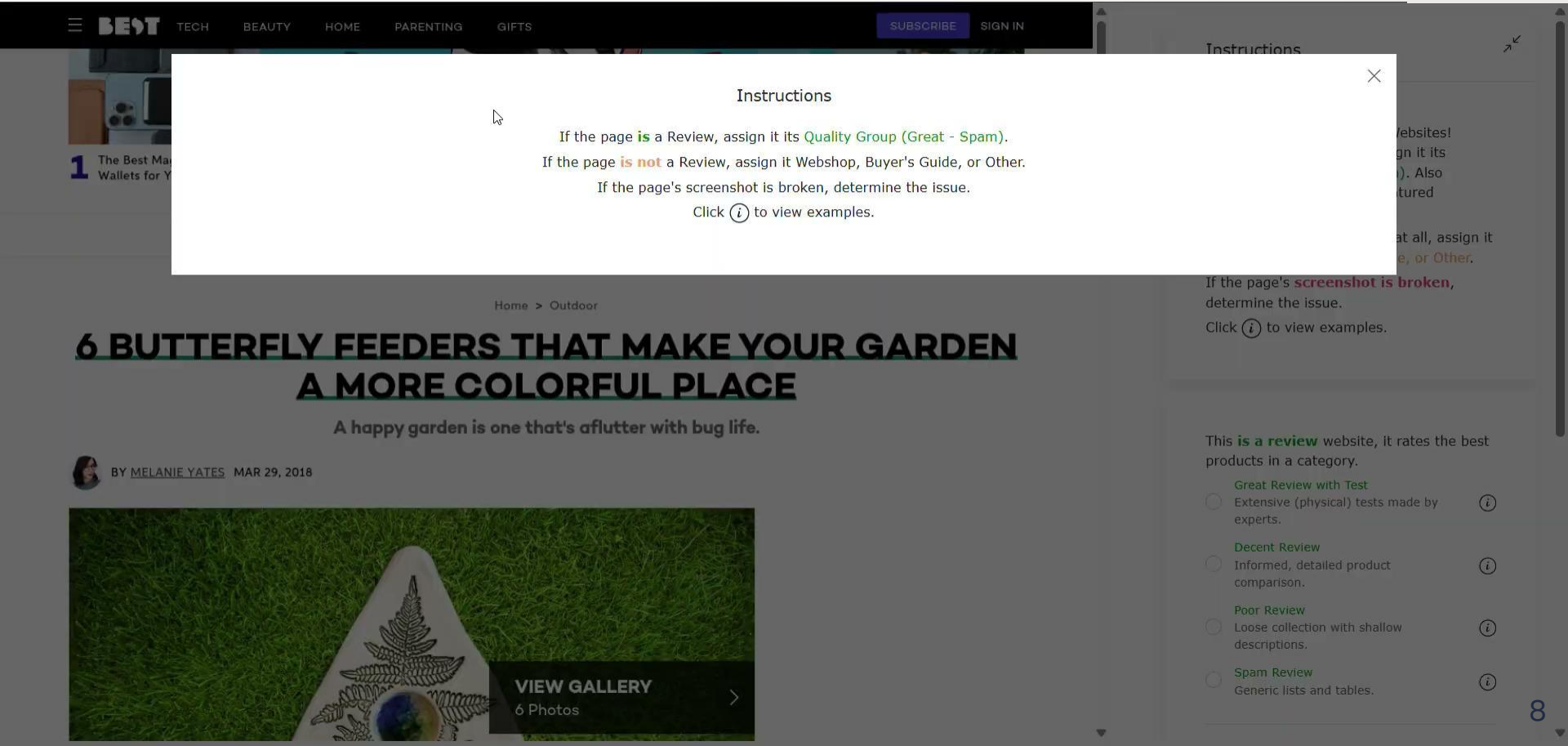
- Used for annotations

*Web ARChive file format is used for storing web crawls and digital archives

Webis Affiliate-Search Corpus



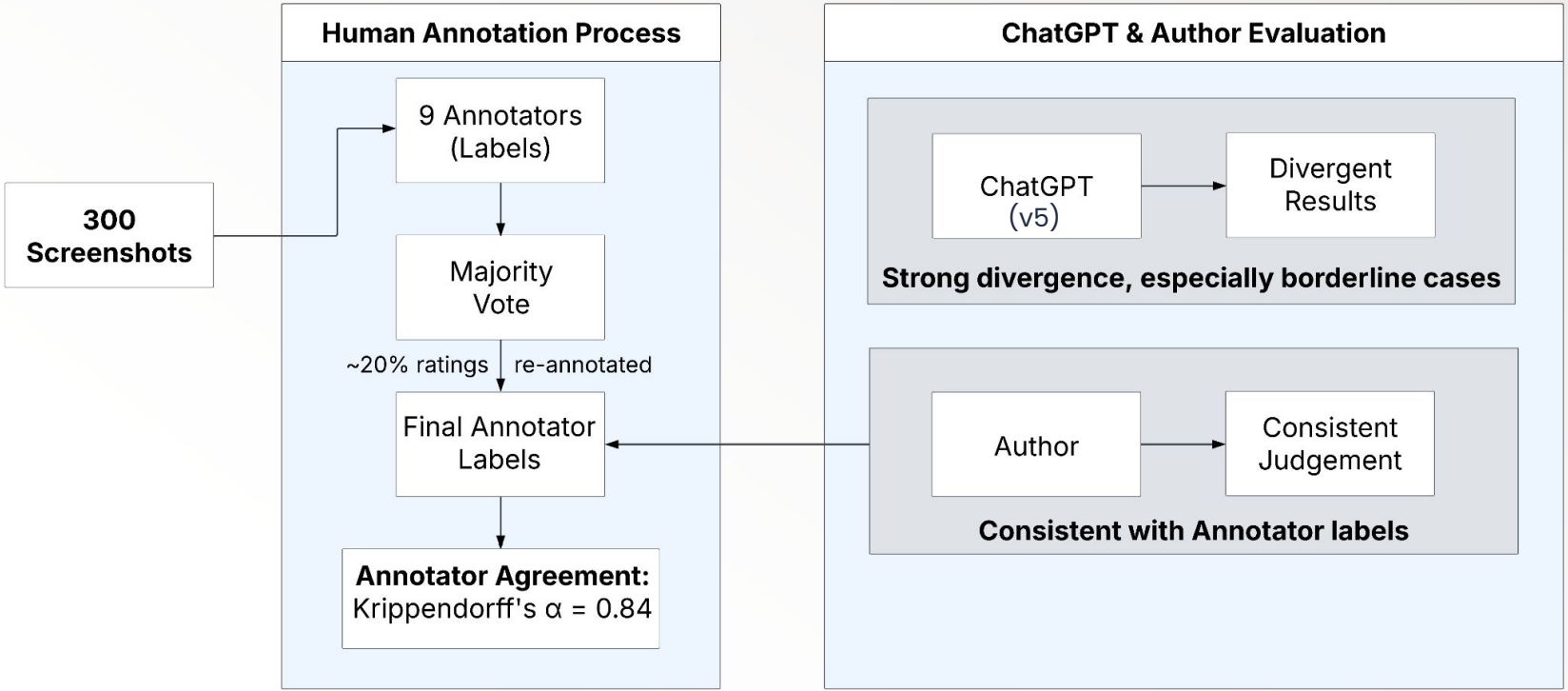
Built Annotation Interface



Initial Annotation Phase (Pilot Study)

- **Initial category labels and instructions were adopted** from an existing work to build the custom annotation interface
- Conducted a pilot study with **3 human annotators**
- Annotated **100 webpage screenshots** across **3 pilot runs**
- Identified ambiguous labels, confusing borderline cases and usability issues
- **Updated** the interface based on **annotator's feedback**
- Refined **category definitions and annotation instructions** iteratively
- Pilot annotations were used only for **task validation, not for model training**

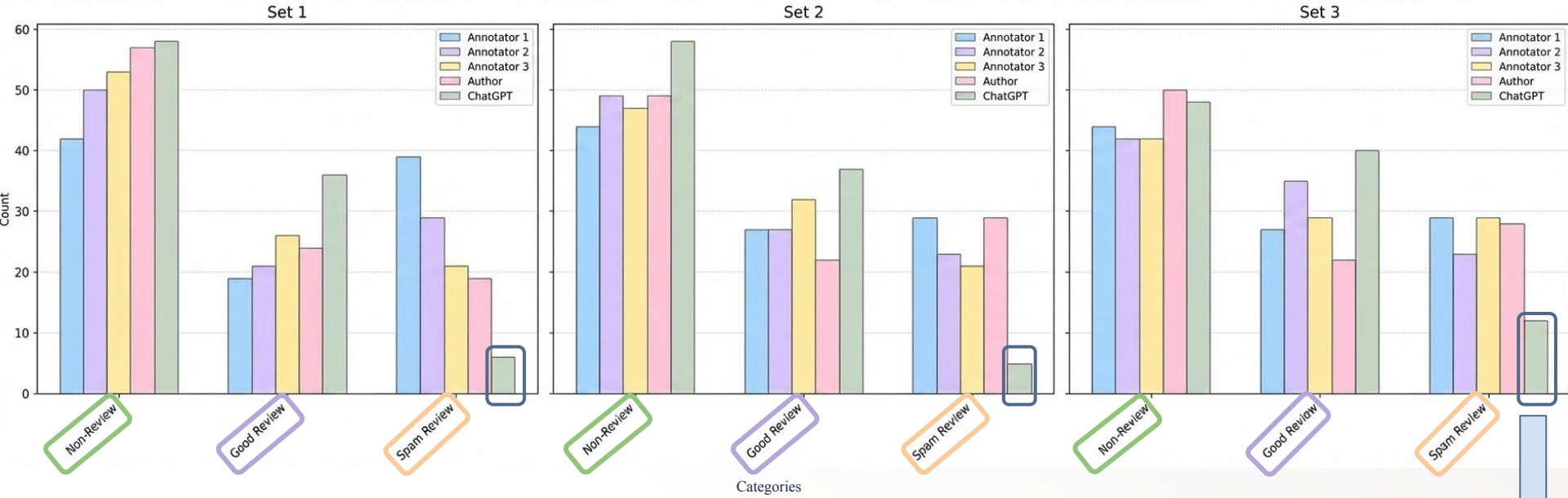
Annotation Task & Agreement



Krippendorff's Alpha measures inter-annotator agreement and a value of 0.84 indicates strong, reliable annotations.

Annotation Results

Figure: This graph shows how different annotators distributed web pages across the **three categories**



(Great+Decent) = Good Review
(Poor+Spam) = Spam Review

Strong *human consistency* in rating, while ChatGPT *misclassified* many spam cases.

PAN Dataset† – Ground truth for AI Authorship Detection

What the PAN Dataset Contains

- **Paired Documents:** 1 Human + 1 AI
- **Source:** 2021 U.S. News Articles
- **Human:** 1,359 Google News Articles
- **AI:** Using the summaries of news articles GPT-4 Turbo created AI text

Why PAN is used in this Thesis

- **Verified Human** vs. **AI Text**
- For **Training & Evaluation**
- **Independent** from Web Data

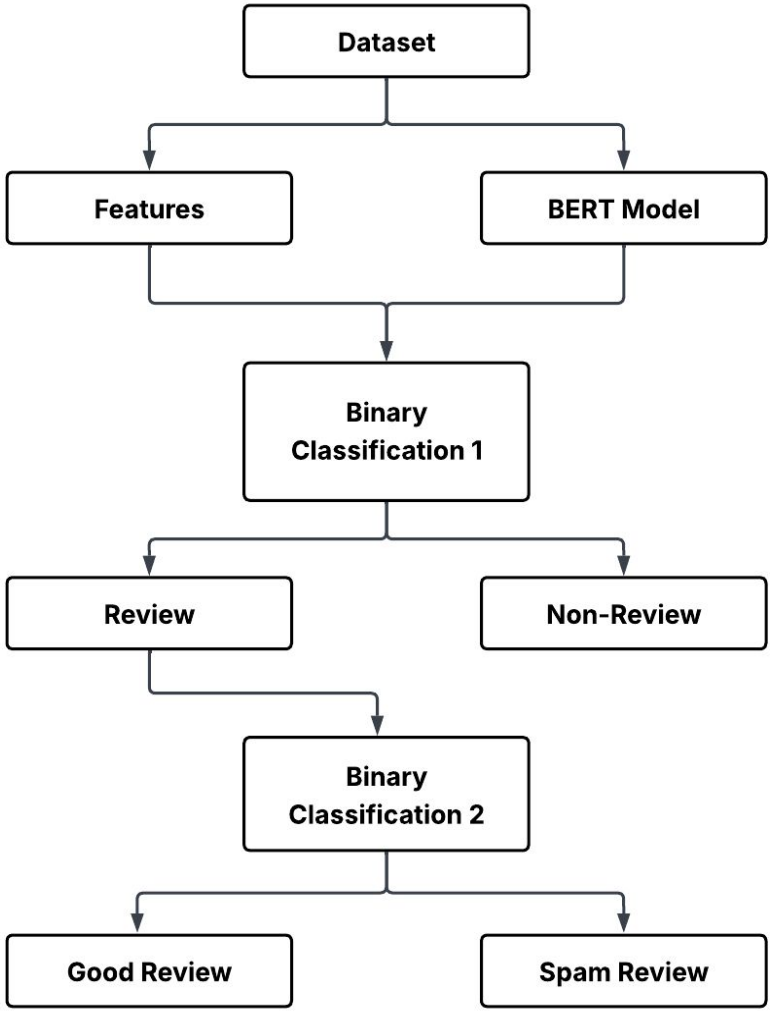
†Janek Bevendorff, Benno Stein, Martin Potthast, et al. Overview of the “voight-kampff” generative ai authorship verification task at pan and eloquent. In CLEF 2024 – Conference and Labs of the Evaluation Forum, CEUR Workshop Proceedings, 2024a. URL https://webis.de/ publications.html#bevendorff_2024d.

Proposed Method: Spam vs Good Review Detection

- Trained on **300** manually annotated webpage screenshots.
- Next, used **DistilBERT** for semantic text understanding. Features: **TF-IDF, POS tags, readability, sentiment**, **subjectivity, length and punctuation density.**
- Two-stage pipeline:

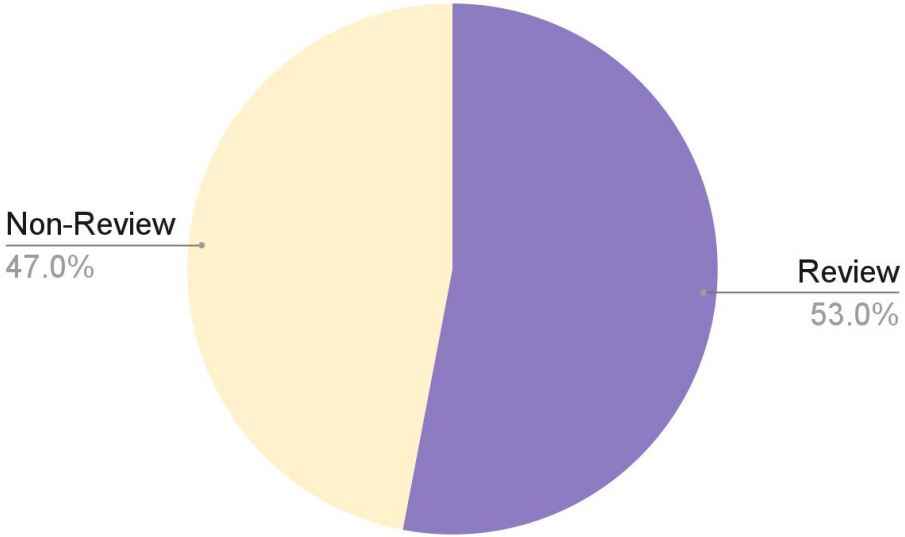
  Review detection → Spam detection
- Final output:

  Good Review or Spam Review
- Performed on ~360,000 pages.

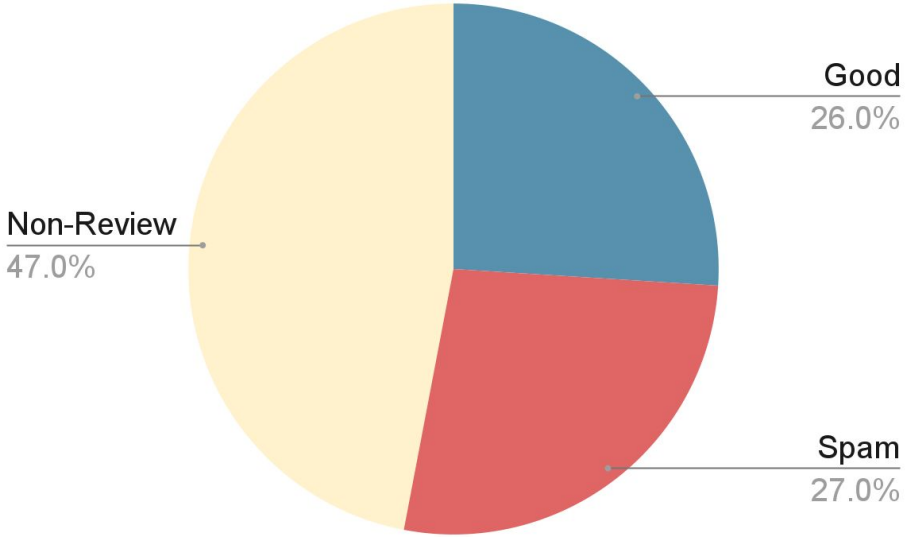


Class Distribution of Annotated Dataset

STAGE 1 : Review Detection (Review vs Non-Review)



STAGE 2 : Review Quality Detection (Good vs Spam)



RESULTS : Spam Detection

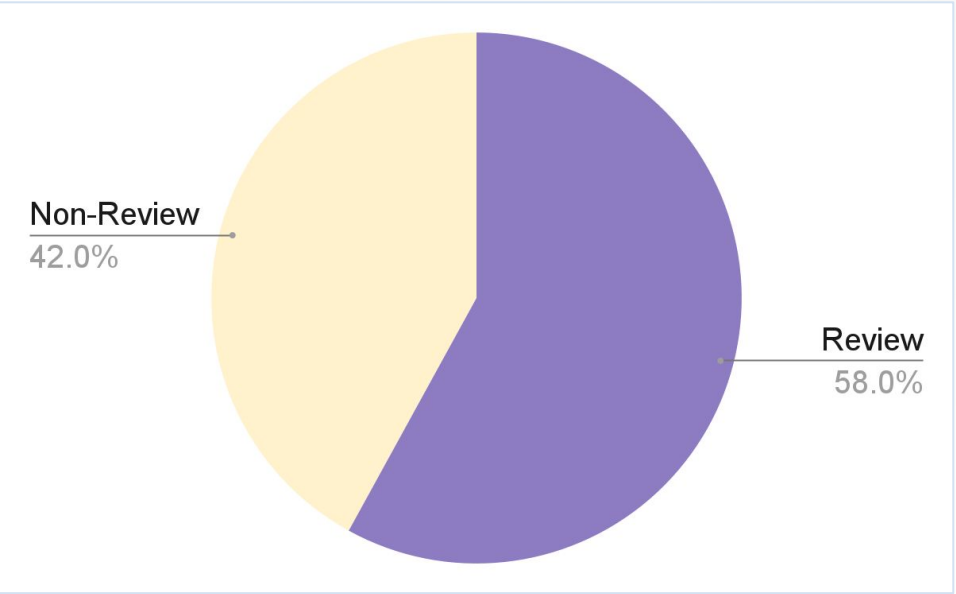
Model	Task	Accuracy
BERT	Binary (Review vs. Non-Review)	0.74
	Binary (Good vs. Spam)	0.55
DistilBERT	Binary (Review vs. Non-Review)	0.78
	Binary (Good vs. Spam)	0.63
RoBERTa	Binary (Review vs. Non-Review)	0.72
	Binary (Good vs. Spam)	0.59

Best Performing Model: DistilBERT

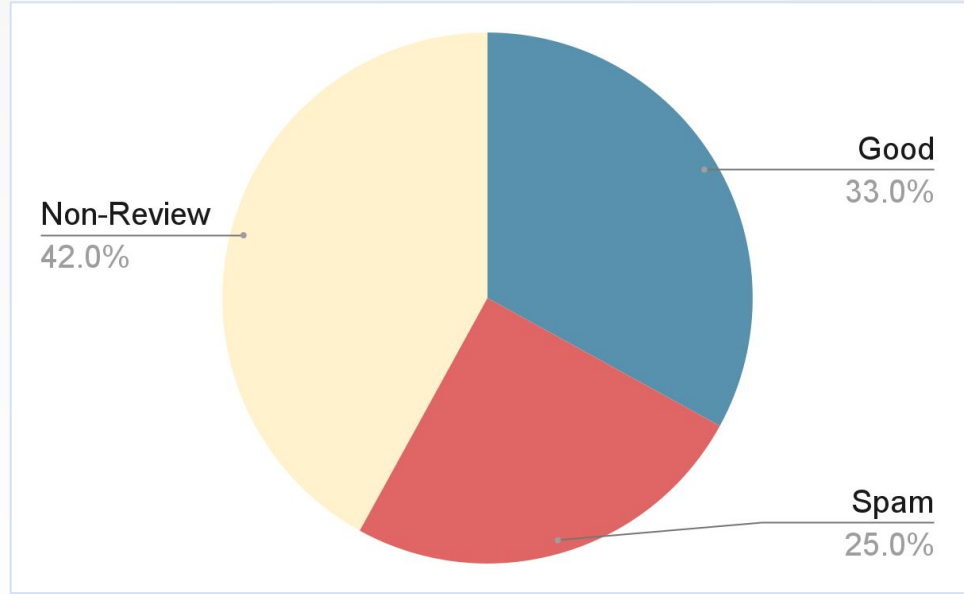
Combining BERT with handcrafted features balances semantic understanding with linguistic signals, improving robustness and stability on noisy review pages.

Class Distribution of Results

STAGE 1 : Review Detection (Review vs Non-Review)

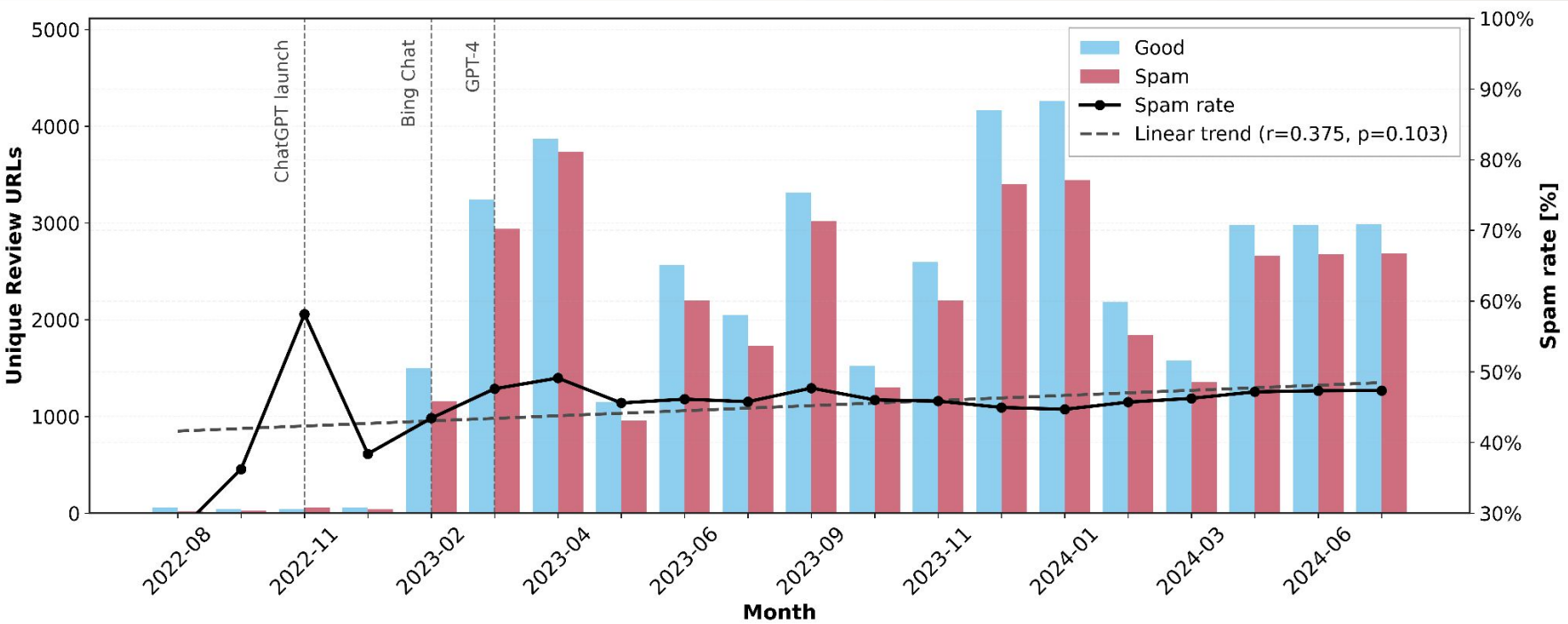


STAGE 2 : Review Quality Detection (Good vs Spam)



Spam Detection: Bing (1/3)

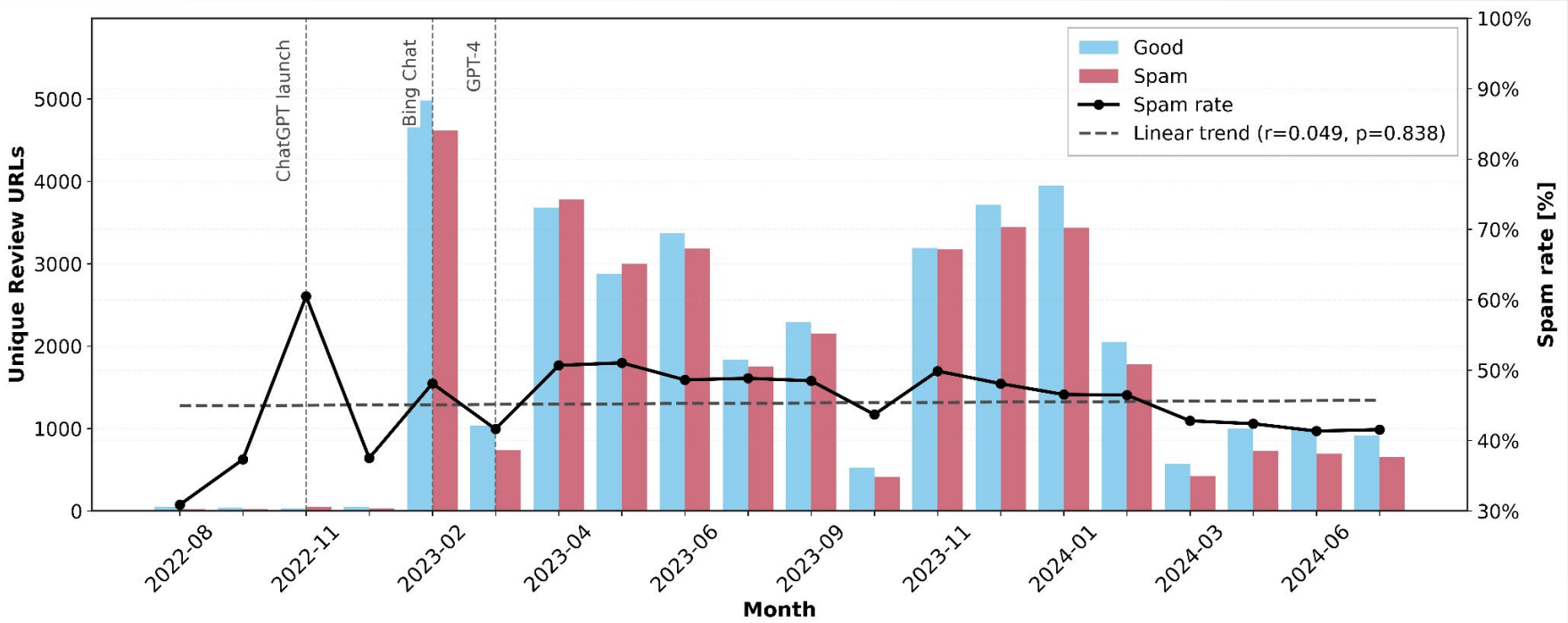
Formula: Spam Rate = n(Spam Reviews) / n(Total Reviews) × 100



Key findings : Bing shows short-term fluctuations in spam rates, but no significant long-term trend, with the overall spam ratio remaining broadly stable (r = 0.375).

DuckDuckGo (2/3)

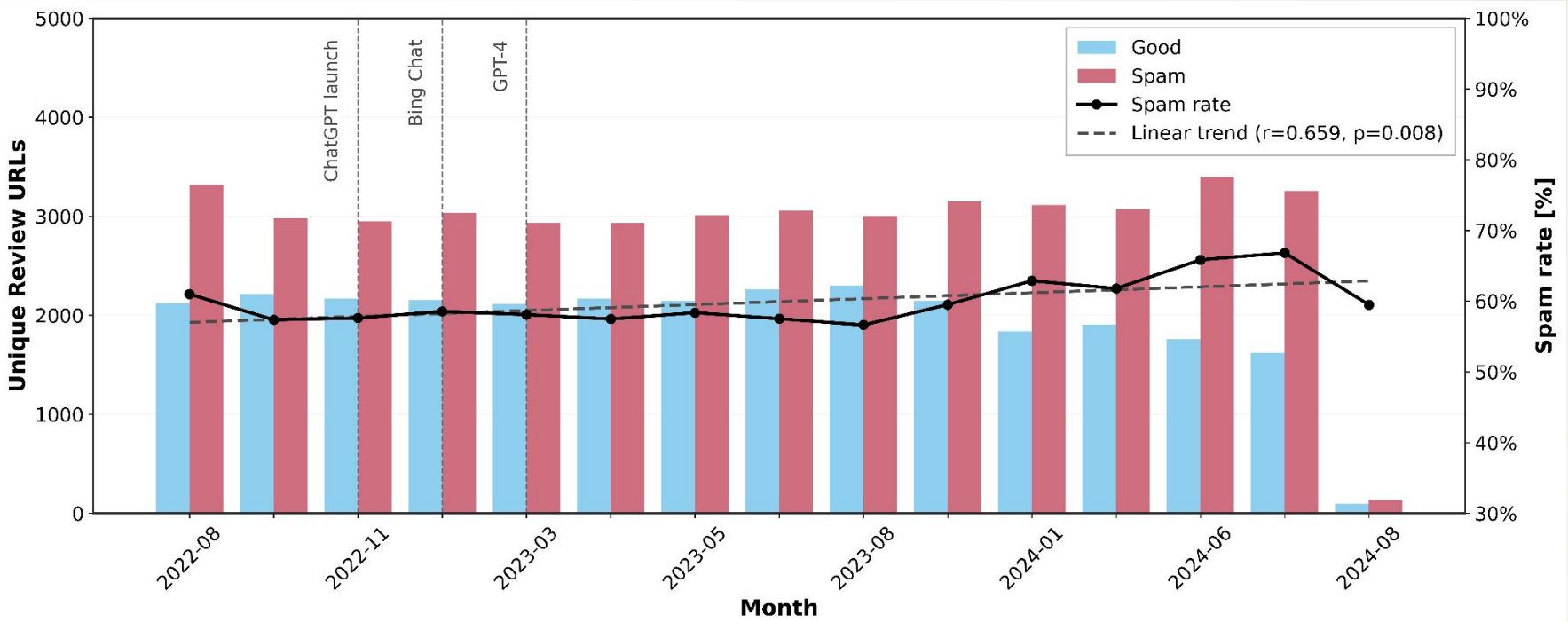
Formula: Spam Rate = n(Spam Reviews) / n(Total Reviews) × 100



Key findings : DuckDuckGo's spam rate shows no meaningful change over time, with an extremely weak and non-significant correlation (r = 0.049).

Startpage (3/3)

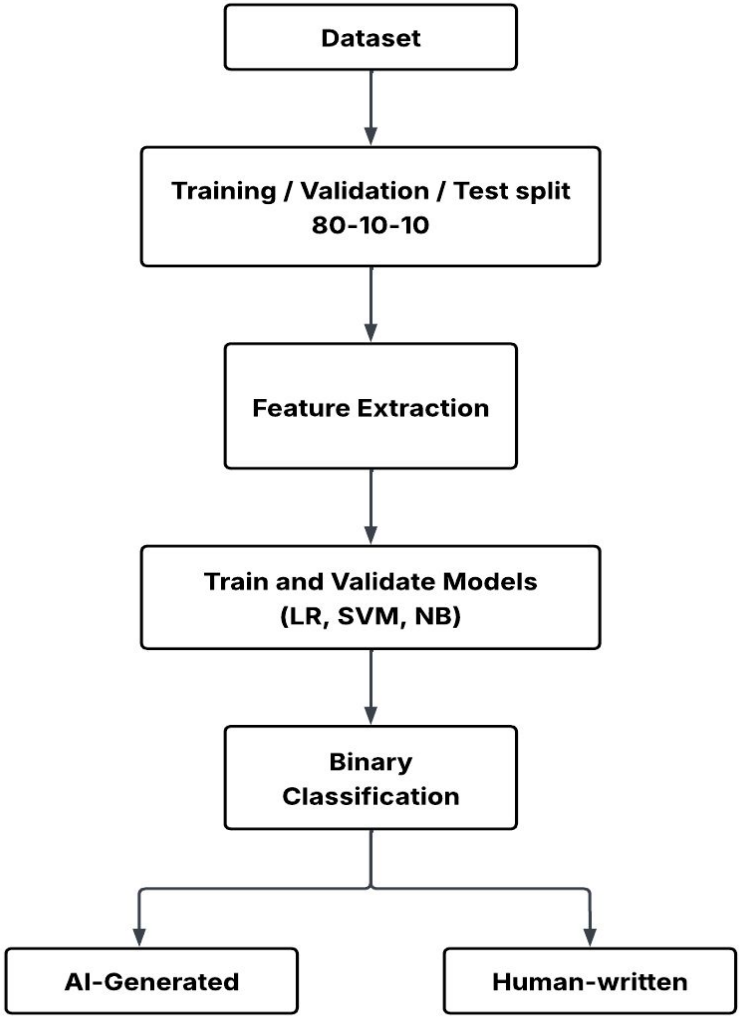
Formula: Spam Rate = n(Spam Reviews) / n(Total Reviews) × 100



Key findings : Startpage shows a moderate, significant increase in spam rates over time, indicating a gradual accumulation of spam content (r = 0.659).

Proposed Method: AI-Generated vs Human Detection

- **PAN dataset** with Human + AI-generated news texts was used
- Model trained on clean, verified Human vs AI texts
- Text converted into **TF-IDF features** (1–2 n-grams)
- **Linear SVM** chosen for high accuracy
- Outputs binary label:
  AI-generated or Human-written
- Performed across 8M+ URLs (whole dataset)



RESULTS : AI Detection

Model Configuration	Accuracy	F1
Support Vector Machine + TF–IDF	0.997	0.997
Support Vector Machine + TF	0.986	0.986
Logistic Regression + TF–IDF	0.990	0.990
Logistic Regression + TF	0.993	0.993
Multinomial Naive Bayes + TF–IDF	0.988	0.988
Multinomial Naive Bayes + TF	0.990	0.990

Best Performing Model: Support Vector Machine + TF–IDF

(Evaluated on PAN Dataset)

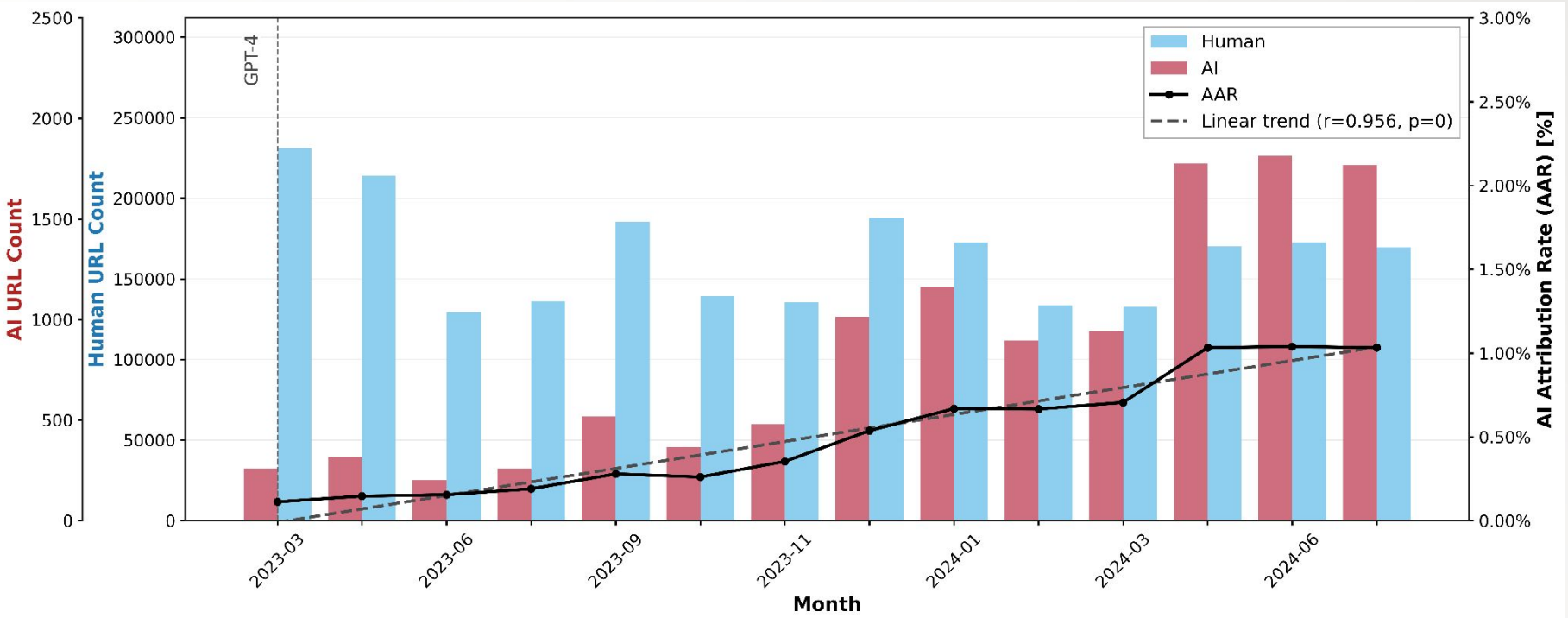
SVM performed better because it maximises class separation in high-dimensional TF–IDF space and is more robust to correlated lexical features than Naive Bayes or Logistic Regression.

Distribution of AI-Generated and Human-Written Content Across Search Engines

Search Engines	Total URLs	AI-Generated	Human	AI Share [%]
Bing + DDG	6,698,886	21,892	6,676,794	0.33
Startpage	1,374,681	5,007	1,369,674	0.36
Total	8,073,367	26,899	8,046,468	0.34

AI Detection: Bing (1/3)

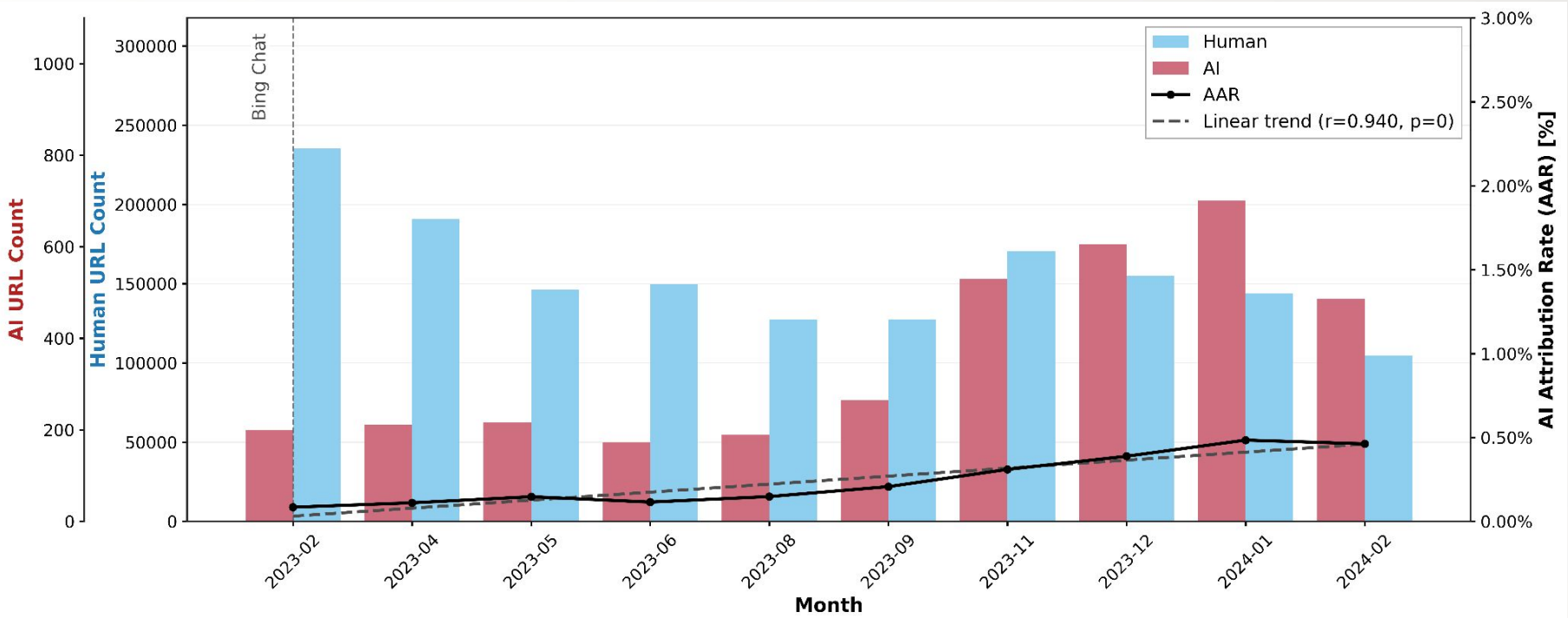
Formula: AI Attribution Rate= n(AI) / n(Total URLs) × 100



Key findings: Bing shows a strong and consistent increase in AI-generated content over time, with a very high correlation to time (r = 0.956), while the overall share remains below 1.1%.

DuckDuckGo (2/3)

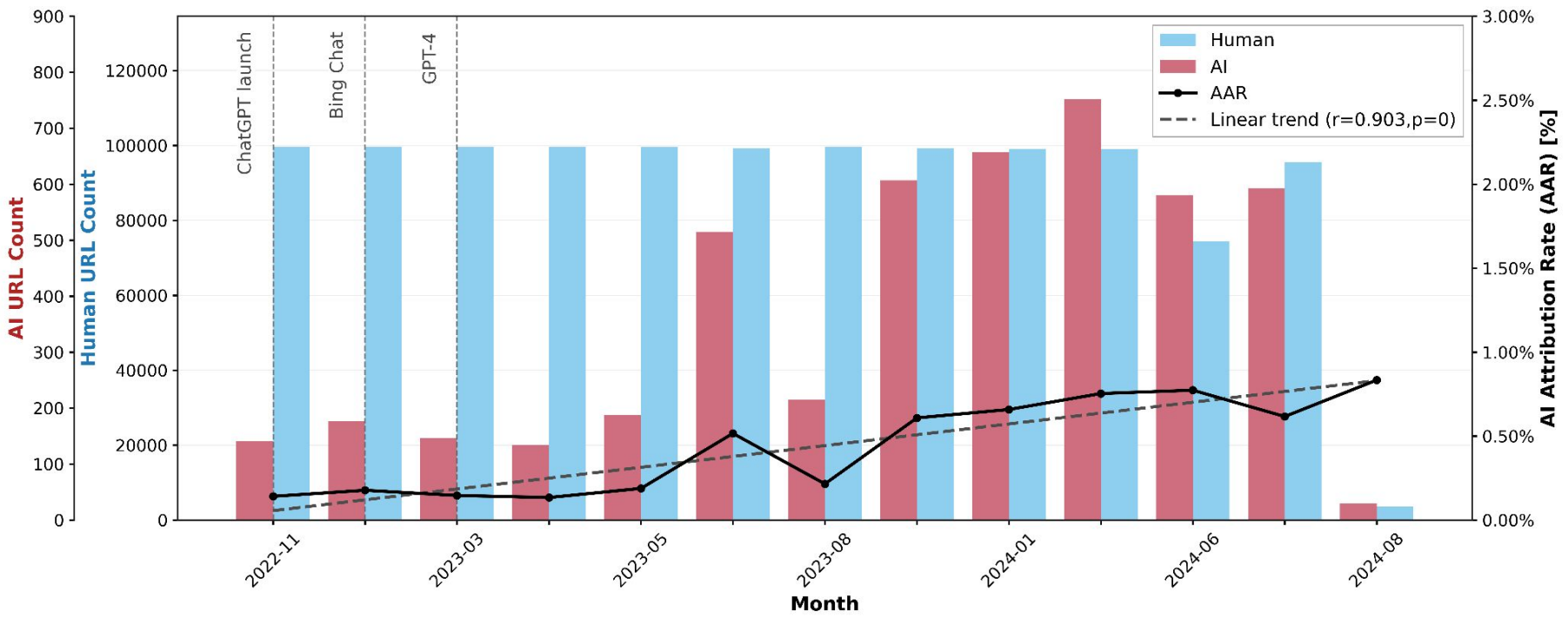
Formula: AI Attribution Rate= n(AI) / n(Total URLs) × 100



Key findings: DuckDuckGo shows a steady increase in AI-generated content over time (r = 0.940), but the overall share remains below 0.5%, with human-written pages continuing to dominate.

Startpage (3/3)

Formula: AI Attribution Rate= n(AI) / n(Total URLs) × 100



Key findings: Startpage shows a gradual but consistent increase in AI-generated content over time (r = 0.903), while human-written pages remain the clear majority below a 1% AI share.

Correlation Analysis – Spam and AI content

- Pearson correlation was used to measure the linear relationship
- Final result: r = 0.022 → **extremely weak correlation**
- AI-generated text and spam reviews **rarely overlap**
- AI text and spam show **different patterns and behaviours**
- Low p-value ($p < 0.001$) shows result is significant, but **practical effect is minimal**

AI text can sound fluent and neutral without trying to promote products, while spam mainly aims to increase visibility using SEO manipulation, so they rarely overlap, resulting in a very weak correlation.

Conclusion

RQ1 — Spam Review Classification

The hybrid DistilBERT model achieved about **78% accuracy in review identification and 63% in spam classification**. This performance shows that even *strong transformer models* struggle when spam closely resembles authentic user feedback.

RQ2 — AI vs Human text Detection

The TF–IDF + Linear SVM (best performer) model *achieved high accuracy 99.7%* and **effectively detected** human-written and AI-generated text across millions of web pages. It is important to note that performance was evaluated against external ground truth.

RQ3 — Correlation Between AI Content and Spam

Correlation analysis showed an extremely **weak relationship (r = 0.022)**, indicating that AI-generated content and spam reviews are *largely independent phenomena* with different patterns and motivations.

RQ4 — Temporal Trends Across Search Engines

AI-generated content and spam **evolved differently** over the **two-year period**.
AI content increased slowly across engines, while spam fluctuations varied significantly by platform, showing distinct behaviours.

THANK YOU!

Do you have any questions?